Lab Manual - LAB 9

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Group A and B

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Bachelor of Science in Software Engineering

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LAB 09: Physical Connectivity and Basic Topologies in Cisco Packet Tracer

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Physical Connectivity and Basic Topologies in Cisco Packet Tracer

Learning Objectives

By the end of this lab, students will be able to:

- Identify common Ethernet cables, connectors, and wiring standards.
- Explain the purpose of straight-through and crossover cables.
- Understand the process of creating, terminating, and testing a LAN cable conceptually.
- Connect two computers in Packet Tracer and test their communication.
- Build and identify point-to-point and bus topologies.
- Explore the same network concepts in the Logical and Physical Workspaces.
- Explain the difference between a network's logical connections and its physical organization.

Introduction to physical connectivity

Physical connectivity refers to the connections that allow network devices to communicate over a transmission medium, such as copper Ethernet cable or fiber-optic cable.

A computer network requires more than IP addresses. Devices also need suitable network interfaces and a connection between them.

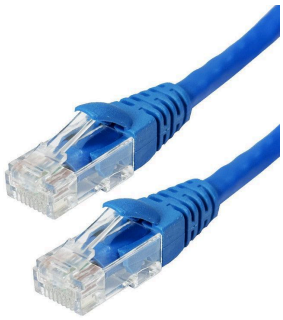
For example, two computers may be configured with valid IP addresses in the same subnet, but they cannot communicate over a wired Ethernet network if there is no working physical connection between them.

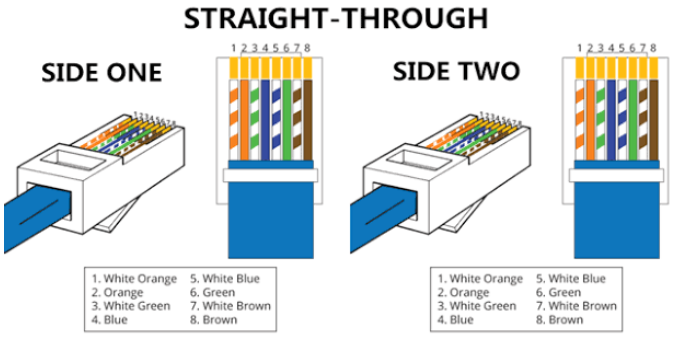
Ethernet cables, connectors, and wiring standards

Ethernet cable types

1. **Straight-through Ethernet cable**

   Commonly used to connect different device types, such as a PC to a switch. Both ends follow the same wiring standard, such as T568B at both ends.

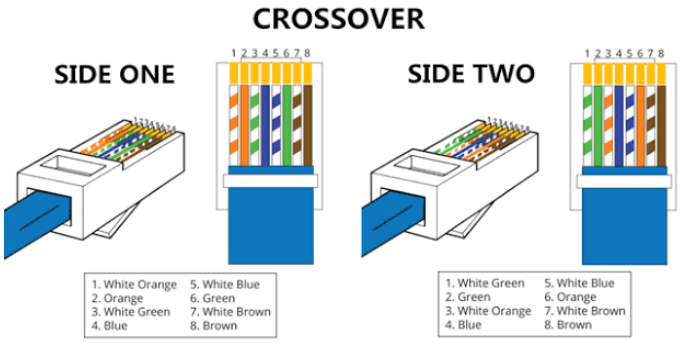




2. **Crossover Ethernet cable**

Traditionally used to connect similar Ethernet devices directly, such as two PCs or two switches. The transmit and receive pairs are crossed.





RJ-45-style Ethernet connector

Ethernet twisted-pair cables commonly use an eight-position modular connector, often called RJ-45 in networking labs. The connector's contacts must match the cable's wiring arrangement.

In a real installation, a technician selects the appropriate cable, arranges its conductors according to a wiring standard, terminates the ends, and tests the cable.

In Packet Tracer, students select a virtual cable and connect device interfaces. The software does not reproduce the complete real-world process of stripping cable jackets, arranging individual wires, crimping connectors, and physically testing a manufactured cable.

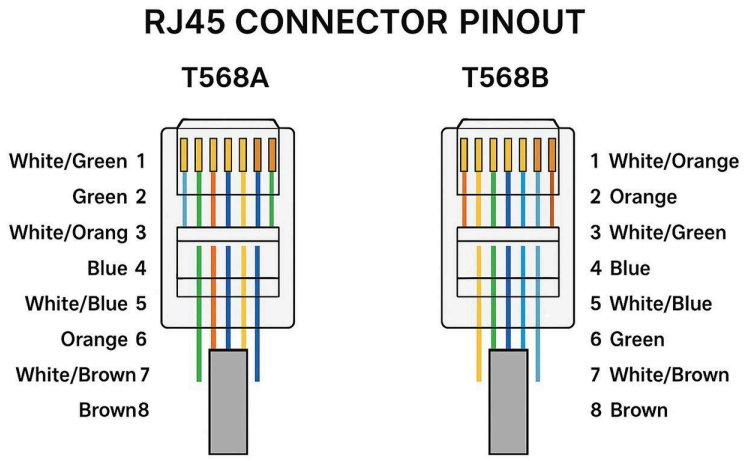


Ethernet wiring standards: T568A and T568B

T568A and T568B define the arrangement of wire pairs at the ends of twisted-pair Ethernet cables.

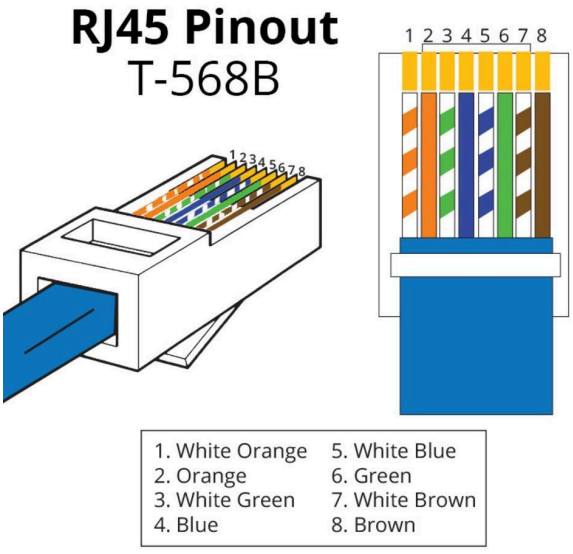
T568A

- One standardized arrangement of the eight conductors.



T568B

- Another standardized arrangement, commonly used in Ethernet patch cabling.



The key distinction is that the green and orange pairs exchange positions between T568A and T568B. The blue and brown pairs remain in the same pin positions.

Pin	T568A	T568B
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

Important: A straight-through cable uses the same wiring standard at both ends. A traditional crossover cable uses T568A at one end and T568B at the other.

Creating, terminating, and testing LAN cables

In the real world, cable creation and termination involve physical tools. Because this lab uses only Cisco Packet Tracer, we will study the process conceptually and practice selecting, connecting, and checking virtual cables.

1. Real-world process

1. Measure and cut the required length of twisted-pair cable.
2. Strip the outer jacket carefully.
3. Arrange the conductors according to T568A or T568B.
4. Trim the conductors evenly.
5. Insert them into the connector in the correct order.
6. Crimp the connector using a compatible tool.
7. Test continuity and wiring using a cable tester.

2. Packet Tracer simulation

1. Open the Logical Workspace.
2. Locate the Connections tool, usually represented by a lightning-bolt icon.
3. Select a suitable Ethernet cable.
4. Click the first device and select its available Ethernet interface.
5. Click the second device and select its interface.
6. Observe the connection line and interface indicators.
7. Wait for the link to become active, where appropriate.
8. Inspect the connection and interface status.

Packet Tracer simulates the connection between device interfaces; it does not physically manufacture, terminate, or certify the cable.

Point-to-Point Topology

A point-to-point topology is a network arrangement in which two endpoints are connected by a dedicated link.



Examples include:

- A direct link between two computers.
- A dedicated link between two routers.
- A point-to-point connection between two network endpoints.

A direct connection between two computers is the simplest example. Point-to-point links are also used in many WAN designs, although the connection technology can differ.

Advantages

- Simple to understand and configure.
- Easy to identify the two endpoints.
- Useful for direct communication and dedicated links.
- Straightforward to troubleshoot when the connection has only two endpoints.

Limitations

- A single link connects only its two endpoints.
- If the link fails, communication over that link is interrupted.
- Connecting many devices with individual point-to-point links can require additional interfaces and cabling.

Practical: Build a point-to-point topology in the Logical Workspace

Connect two PCs directly, configure IPv4 addresses, and verify communication using ping.

Devices required

- PC0
- PC1
- Copper crossover cable, unless the device and Packet Tracer version support direct communication using another suitable connection

Step 1: Add the computers

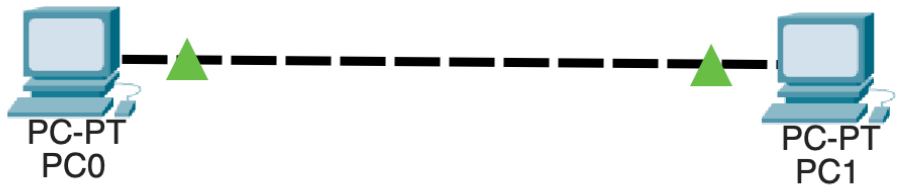
1. Open Cisco Packet Tracer.
2. Select the Logical Workspace.
3. From the End Devices category, add two PCs.
4. Arrange them with sufficient space between them.
5. Rename them PC0 and PC1 if necessary.

Step 2: Connect the computers

1. Select the Connections tool.

2. Select Copper Cross-Over.
3. Click PC0 and select its FastEthernet interface.
4. Click PC1 and select its FastEthernet interface.
5. Wait for the link indicators to settle.

A crossover cable is the traditional choice for a direct PC-to-PC Ethernet connection. If your version or simulated interfaces support automatic crossover, a different cable may also establish the link.



Step 3: Configure IPv4 addresses

Open PC0, select Desktop, and open IP Configuration. Use the following settings:

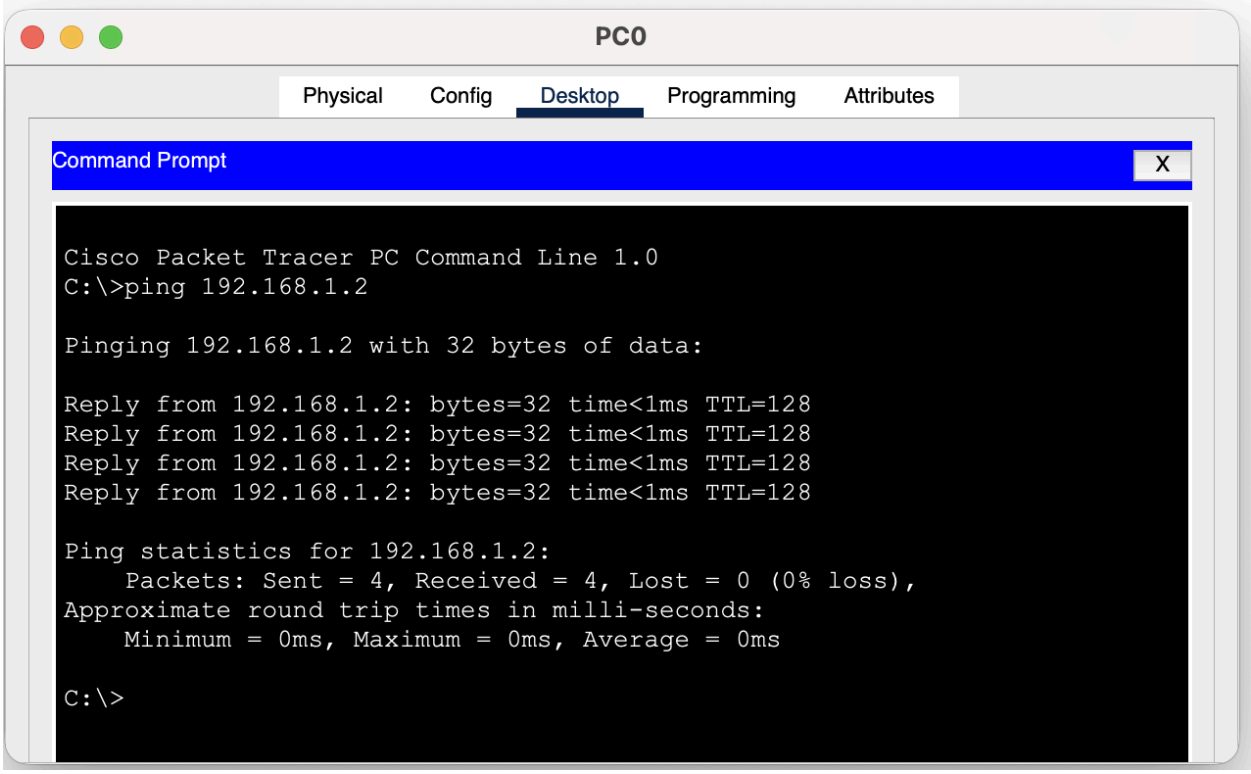
Setting	PC0	PC1
IPv4 address	192.168.1.1	192.168.1.2
Subnet mask	255.255.255.0	255.255.255.0

Both computers are in the same subnet, so a router or default gateway is not required for this direct communication test.

Step 4: Test communication

On PC0:

1. Open Desktop.
2. Select Command Prompt.
3. Enter: ping 192.168.1.2



Step 5: Troubleshooting

If the ping fails, check the following:

- Is the Ethernet cable connected to both PCs?
- Are the correct interfaces selected?
- Is the link active?
- Are both IP addresses correctly configured?
- Do both PCs use the same subnet mask?
- Is the selected cable suitable for the simulated connection?

View point-to-point connectivity in the Physical Workspace

The Logical Workspace displays the topology clearly, while the Physical Workspace provides a location-oriented view of network infrastructure.

Procedure

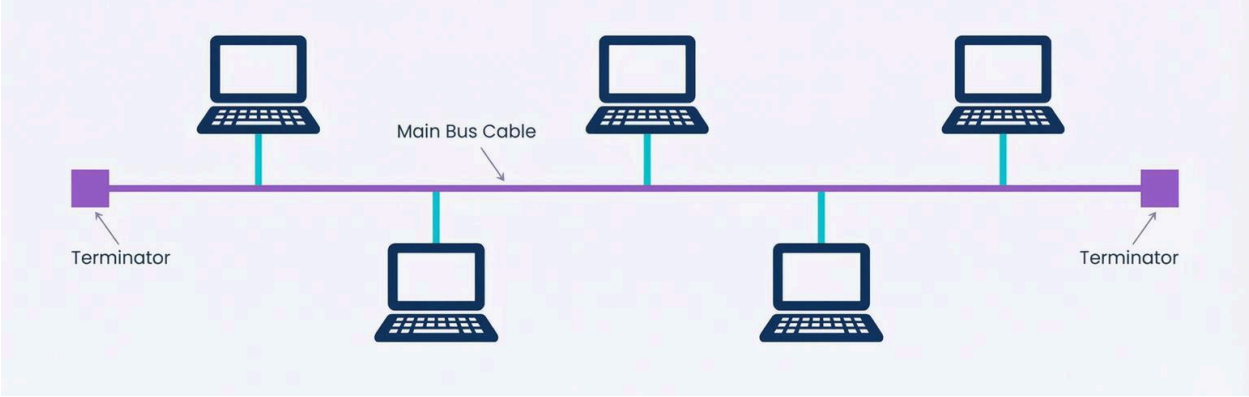
1. Save the point-to-point project.
2. Select the Physical Workspace tab.

3. Explore the physical environment that is available.
4. Locate the devices in the environment, where supported.
5. Observe how the physical view differs from the logical diagram.



Bus Topology and Physical Workspace

In a traditional bus topology, multiple devices share a common communication backbone. Each device attaches to the shared medium, and the medium has two ends.



Key characteristics

- Devices share a common backbone.
- The traditional arrangement uses terminators at the ends of the backbone.

- A fault in the backbone can affect communication for many or all devices.
- Troubleshooting can become difficult as the network grows.
- Classic bus networks were historically associated with coaxial Ethernet technologies.

Advantages

- Simple conceptual structure.
- Historically required less backbone cabling than some alternatives.
- Useful for learning how shared-medium networks operate.

Limitations

- Backbone failure can disrupt the network.
- Fault isolation can be difficult.
- Performance and collision behavior depend on the technology used.
- It is not the usual physical arrangement for modern switched Ethernet LANs.

Important Packet Tracer limitation: Representing a bus

A traditional bus topology uses a shared backbone medium. A normal modern Ethernet switch with several PCs connected to it creates a star-shaped physical topology, not a true bus.

Packet Tracer may not support a traditional coaxial bus arrangement in the same way that it supports modern Ethernet connections.

Practical: Explore a bus-like shared-medium network

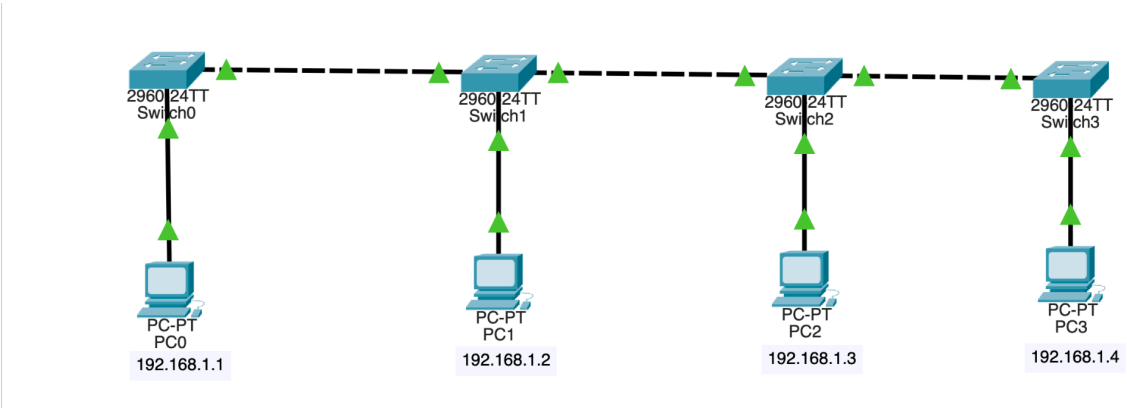
Devices

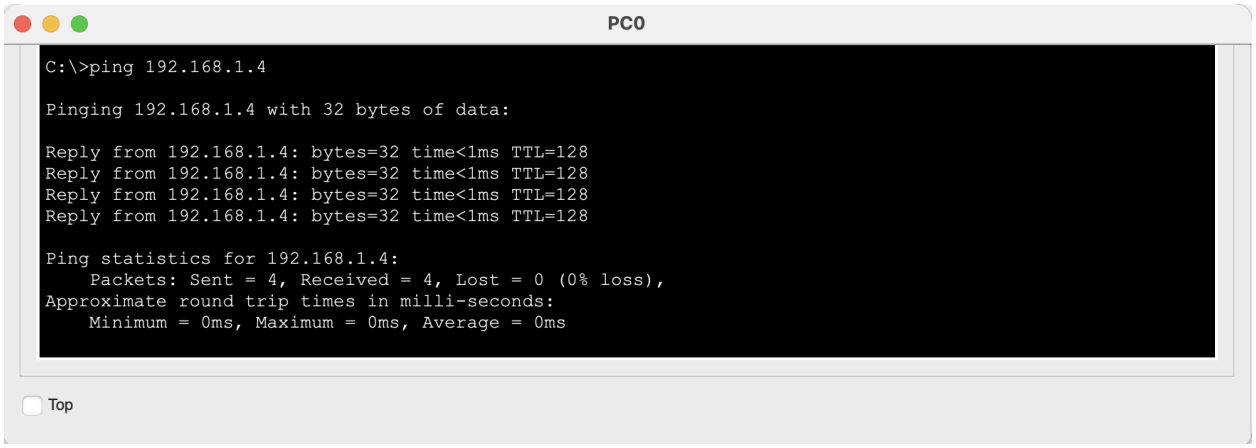
- Four PCs.
- Four Switches.
- Suitable Ethernet connections.
- The Physical Workspace for location-oriented inspection.

Procedure A: Build the supported approximation

1. Open a new Packet Tracer project.
2. Add four PCs and four switches.
3. Connect each PC to each switch using suitable Ethernet cables.
4. Connect switches to each other.
5. Arrange the devices neatly in the Logical Workspace.
6. Configure unique IP addresses in the same subnet.
7. Test connectivity between two PCs using ping.
8. Observe the connection structure.

Device	IPv4 address	Subnet mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0





Explore the Physical Workspace

1. Switch to the Physical Workspace.
2. Inspect the available physical environment and device-placement controls.
3. Explore how the PCs and Switch are represented.
4. If supported, organize devices within an available physical area.
5. Return to the Logical Workspace.



Comparison: Point-to-point vs. bus topology

Feature	Point-to-point	Classic bus
Structure	One link between two endpoints	Multiple devices share a backbone
Number of endpoints	Two per point-to-point link	Multiple devices
Typical example	Direct link between two PCs	Historical coaxial Ethernet LAN
Main advantage	Simple and easy to troubleshoot	Simple shared-medium concept
Main limitation	One link connects only two endpoints	Backbone failure can affect many devices
Packet Tracer demonstration	Direct connection between two PCs	Conceptual diagram or supported shared-medium approximation